

WEEK – 2

ENTITY ANALYSIS REPORT

Project Title

ShopNest – E-Commerce Order
Management Database System

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1. Introduction

The **ShopNest – E-Commerce Order Management Database System** is designed to manage the daily operations of an online shopping platform using a centralized relational database. The system stores and manages information related to customers, product categories, products, suppliers, customer orders, payments, shipments, and customer reviews.

Entity analysis is one of the most important phases of database design. It identifies the data objects required by the system, their attributes, and the constraints needed to maintain data accuracy and consistency. A well-defined entity structure minimizes data redundancy, improves data integrity, and supports efficient business operations.

The objective of this report is to identify the entities required for the system, define their attributes, specify the primary keys, foreign keys, and database constraints that will be used in the database design.

2. Entity Identification

Based on the Software Requirement Specification (SRS), the following entities are identified for the ShopNest E-Commerce Order Management Database System.

Entity	Purpose
Customer	Stores customer information.
Category	Stores product category information.
Product	Stores product details.
Supplier	Stores supplier information.
Order	Stores customer order information.
Order Details	Stores the products included in each order.
Payment	Stores payment information.
Shipment	Stores shipment and delivery information.
Review	Stores customer ratings and comments.

3. Entity Attribute List

Customer

Attribute	Description
Customer_ID	Unique customer identifier
Name	Customer name
Email	Customer email address
Mobile_Number	Customer mobile number
Address	Customer address
Registration_Date	Registration date

Category

Attribute	Description
Category_ID	Unique category identifier
Category_Name	Category name
Description	Category description

Product

Attribute	Description
Product_ID	Unique product identifier
Product_Name	Product name
Price	Product price
Stock_Quantity	Available stock
Category_ID	Category reference

Supplier

Attribute	Description
Supplier_ID	Unique supplier identifier
Supplier_Name	Supplier name

Contact_Information	Supplier contact details
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Order

Attribute	Description
Order_ID	Unique order identifier
Customer_ID	Customer reference
Order_Date	Order date
Total_Amount	Total order amount
Order_Status	Current order status

Order Details

Attribute	Description
Order_Detail_ID	Unique order detail identifier
Order_ID	Order reference
Product_ID	Product reference
Quantity	Ordered quantity
Unit_Price	Price per product

Payment

Attribute	Description
Payment_ID	Unique payment identifier
Order_ID	Order reference
Payment_Method	Payment method
Payment_Date	Payment date
Payment_Status	Payment status

Shipment

Attribute	Description
Shipment_ID	Unique shipment identifier
Order_ID	Order reference
Delivery_Address	Delivery address
Shipment_Date	Shipment date
Delivery_Status	Delivery status

Review

Attribute	Description
Review_ID	Unique review identifier
Customer_ID	Customer reference
Product_ID	Product reference
Rating	Product rating
Comments	Customer comments

4. Primary Key Identification

Entity	Primary Key
Customer	Customer_ID
Category	Category_ID
Product	Product_ID
Supplier	Supplier_ID
Order	Order_ID
Order Details	Order_Detail_ID
Payment	Payment_ID
Shipment	Shipment_ID
Review	Review_ID

5. Foreign Key Identification

Entity	Foreign Key
Product	Category_ID
Order	Customer_ID
Order Details	Order_ID, Product_ID
Payment	Order_ID
Shipment	Order_ID
Review	Customer_ID, Product_ID

6. Database Constraints

Customer

Attribute	Constraint
Customer_ID	PRIMARY KEY, NOT NULL
Name	NOT NULL
Email	NOT NULL, UNIQUE
Mobile_Number	NOT NULL, UNIQUE
Address	NOT NULL
Registration_Date	NOT NULL, DEFAULT CURRENT_DATE

Category

Attribute	Constraint
Category_ID	PRIMARY KEY, NOT NULL
Category_Name	NOT NULL, UNIQUE
Description	NULL Allowed

Product

Attribute	Constraint
Product_ID	PRIMARY KEY, NOT NULL
Product_Name	NOT NULL
Price	NOT NULL, CHECK (Price > 0)
Stock_Quantity	NOT NULL, DEFAULT 0, CHECK (Stock_Quantity >= 0)
Category_ID	FOREIGN KEY, NOT NULL

Supplier

Attribute	Constraint
Supplier_ID	PRIMARY KEY, NOT NULL
Supplier_Name	NOT NULL
Contact_Information	NOT NULL

Order

Attribute	Constraint
Order_ID	PRIMARY KEY, NOT NULL
Customer_ID	FOREIGN KEY, NOT NULL
Order_Date	NOT NULL, DEFAULT CURRENT_DATE
Total_Amount	NOT NULL, CHECK (Total_Amount >= 0)
Order_Status	NOT NULL, DEFAULT 'Pending'

Order Details

Attribute	Constraint
Order_Detail_ID	PRIMARY KEY, NOT NULL
Order_ID	FOREIGN KEY, NOT NULL
Product_ID	FOREIGN KEY, NOT NULL
Quantity	NOT NULL, CHECK (Quantity > 0)
Unit_Price	NOT NULL, CHECK (Unit_Price > 0)

Payment

Attribute	Constraint
Payment_ID	PRIMARY KEY, NOT NULL
Order_ID	FOREIGN KEY, NOT NULL
Payment_Method	NOT NULL
Payment_Date	NOT NULL, DEFAULT CURRENT_DATE
Payment_Status	NOT NULL, DEFAULT 'Pending'

Shipment

Attribute	Constraint
Shipment_ID	PRIMARY KEY, NOT NULL
Order_ID	FOREIGN KEY, NOT NULL
Delivery_Address	NOT NULL
Shipment_Date	NOT NULL, DEFAULT CURRENT_DATE
Delivery_Status	NOT NULL, DEFAULT 'Processing'

Review

Attribute	Constraint
Review_ID	PRIMARY KEY, NOT NULL
Customer_ID	FOREIGN KEY, NOT NULL
Product_ID	FOREIGN KEY, NOT NULL
Rating	NOT NULL, CHECK (Rating BETWEEN 1 AND 5)
Comments	NULL Allowed

7. Summary

The entity analysis identified **nine core entities** required for the ShopNest E-Commerce Order Management Database System. Each entity has clearly defined attributes, primary keys, foreign keys, and database constraints. These entities provide the foundation for managing customer information, products, categories, suppliers, orders, payments, shipments, and reviews while maintaining data integrity and minimizing redundancy.

8. Conclusion

The **Entity Analysis Report** defines the data requirements of the ShopNest – E-Commerce Order Management Database System by identifying all required entities, attributes, primary keys, foreign keys, and database constraints. This analysis forms the basis for the next phase of the project, which includes Entity-Relationship (ER) modeling, relationship analysis, cardinality analysis, normalization, and SQL database implementation. A well-structured entity design ensures data consistency, efficient storage, and reliable database performance.